

JAVA Programs

// P1) print AliaBhatt

```
class HelloProg{  
    public static void main(String[] args){  
        System.out.println("Hello..! AliaBhatt");  
    }  
}
```

output:

Hello..! AliaBhatt

//P2) Addition of 2 numbers

```
import java.util.Scanner;  
  
class Addition{  
    public static void main(String[] args){  
        int a,b,c;  
        Scanner s=new Scanner(System.in);  
        System.out.println("Enter a & b values: ");  
        a=s.nextInt();  
        b=s.nextInt();  
        c=a+b;  
        System.out.println("Sum of two numbers:"+c);  
    }  
}
```

output:

Enter a & b values:

56

6

Sum of two numbers:62

//P3) Arithmetic

```
import java.util.Scanner;

class Arithmetic{

public static void main(String[] args){

int a,b,sum,sub,mul,div,mod;

Scanner s=new Scanner(System.in);

System.out.println("Enter a & b values: ");

a=s.nextInt();

b=s.nextInt();

sum=a+b;

System.out.println("Sum of two numbers:"+sum);

sub=a-b;

System.out.println("Subraction of two numbers:"+sub);

mul=a*b;

System.out.println("Multiplication of two numbers:"+mul);

div=a/b;

System.out.println("Division of two numbers:"+div);

mod=a%b;

System.out.println("Modulus of two numbers:"+mod);

}

}
```

Output:

Enter a & b values:

4 8

Sum of two numbers:12

Subsraction of two numbers:-4

Multiplication of two numbers:32

Division of two numbers:0

Modulus of two numbers:4

//P4) swapping of two numbers using third variable

```
import java.util.Scanner;
```

```

class Swap{
public static void main(String[] args){
int a,b,temp;
Scanner s=new Scanner(System.in);
System.out.println("Enter a & b values: ");
a=s.nextInt();
b=s.nextInt();
System.out.println("Before swapping a= "+a+" b= "+b);
temp=b;
b=a;
a=temp;
System.out.println("After swapping a= "+a+" b= "+b);
}
}

```

output:

Enter a & b values:

5 9

Before swapping a= 5 b= 9

After swapping a= 9 b= 5

//P5)swapping of two numbers without third variable

```

import java.util.Scanner;

class Swap{
public static void main(String[] args){
int a,b;
Scanner s=new Scanner(System.in);
System.out.println("Enter a & b values: ");
a=s.nextInt();
b=s.nextInt();
System.out.println("Before swapping a= "+a+" b= "+b);
a=(a+b)-(b=a);
System.out.println("After swapping a= "+a+" b= "+b);
}
}

```

```
}
```

```
}
```

output:

Enter a & b values:

5 8

Before swapping a= 5 b= 8

After swapping a= 8 b= 5

//P6) Check given number is Even or Odd

```
import java.util.Scanner;
```

```
class EvenorOdd{
```

```
public static void main(String[] args){
```

```
int a;
```

```
Scanner s=new Scanner(System.in);
```

```
System.out.println("Enter a number: ");
```

```
a=s.nextInt();
```

```
if(a%2==0){
```

```
System.out.println(a+" is Even");
```

```
}
```

```
else{
```

```
System.out.println(a+" is Odd");
```

```
}
```

```
}
```

```
}
```

output:

Enter a number:

69

69 is Odd

//P7) Check given number is positive or not

```
import java.util.Scanner;
```

```
class PositiveOrNegative{
```

```
public static void main(String[] args){
```

```
int a;
Scanner s=new Scanner(System.in);
System.out.println("Enter a number: ");
a=s.nextInt();
if(a>0){
System.out.println(a+" is Positive");
}
else{
System.out.println(a+" is Negative");
}
}
}
```

output:

Enter a number:

-98

-98 is Negative

//P8) sum of even and odd numbers in given range

```
import java.util.Scanner;
class PositiveOrNegative{
public static void main(String[] args){
int min,max,oddSum=0,evenSum=0;
Scanner s=new Scanner(System.in);
System.out.println("Enter minimum and maximum values: ");
min=s.nextInt();
max=s.nextInt();
for(int i=min;i<=max;i++){
if(i%2!=0){
oddSum+=i;
}
else{
evenSum+=i;
}
```

```
}  
}  
System.out.println("Sum of Odd numbers: "+oddSum+"\nSum of Even Sum: "+evenSum);  
}  
}
```

output:

Enter minimum and maximum values:

1 10

Sum of Odd numbers: 25

Sum of Even Sum: 30

//P9) Reverse a number

```
import java.util.Scanner;  
class Reverse{  
    public static void main(String[] args){  
        int num,rev=0;  
        Scanner s=new Scanner(System.in);  
        System.out.println("Enter number: ");  
        num=s.nextInt();  
        while(num>0){  
            int rem=num%10;  
            rev=rev*10+rem;  
            num=num/10;  
        }  
        System.out.println("After: "+rev);  
    }  
}
```

output:

Enter number:

5678

After: 8765

//P10) Check given number is prime or not

```

import java.util.Scanner;

class PrimeNumber{

public static void Prime(int n){

int count=0;

for(int i=1;i<=n;i++){

if(n%i==0)

count++;

}

if(count==2){

System.out.println(n+" is a Prime number.");

}

else{

System.out.println(n+" is not a Prime number.");

}

}

public static void main(String[] args){

int num;

Scanner s=new Scanner(System.in);

System.out.println("Enter number: ");

num=s.nextInt();

Prime(num);

}

}

```

outrput:

Enter number:

17

17 is a Prime number.

//P11) Check given number is prime or not and count the numbers in the given range

```

import java.util.Scanner;

class PrimeNumber{

public static boolean Prime(int n){

```

```

if(n<=1)
return false;
for(int i=2;i<n;i++){
if(n%i==0)
return false;
}
return true;
}

public static void main(String[] args){
int num,count=0;
Scanner s=new Scanner(System.in);
System.out.println("Enter number: ");
num=s.nextInt();
for(int i=0;i<=num;i++){
if(Prime(i)){
count++;
System.out.print(i+" ");
}
}

System.out.println("\nCount of prime numbers in the given rage: "+count);
}
}

```

output:

Enter number:

15

2 3 5 7 11 13

Count of prime numbers in the given rage: 6

//P12) To check given number is Perfect Number or not

```

import java.util.Scanner;

class PerfectNum{

public static void main(String[] args){

```



```
int num,sum=0;
Scanner s=new Scanner(System.in);
System.out.println("Enter number: ");
num=s.nextInt();
for(int i=1;i<=num/2;i++){
    if(num%i==0){
        sum+=i;
    }
}
if(num==sum){
    System.out.println(sum+" is perfect number");
}
else{
    System.out.println(sum+" is not perfect number");
}
}
```

output:

Enter number:

496

496 is perfect number

//P13) Using Quadratic equation to find roots

```
import java.util.Scanner;

class QuadraticEquation{

    public static void main(String[] args){

        double a,b,c,d,r1,r2;

        Scanner s=new Scanner(System.in);

        System.out.println("Enter a, b & c values: ");

        a=s.nextDouble();

        b=s.nextDouble();

        c=s.nextDouble();
```

```

d=(b*b)-(4*a*c);
if(d>0){
r1=(-b+Math.sqrt(d))/(2*a);
r2=(-b-Math.sqrt(d))/(2*a);
System.out.printf("r1: %.4f r2: %.4f are real & different.",r1,r2);
}
else if(d==0){
r1=r2=-b/(2*a);
System.out.println("r1: "+r1+"r2:" + r2+" are real & same.");
}
else{
System.out.println("Roots are imaginary.");
}
}
}
}

```

output:

Enter a, b & c values:

1 4 1

r1: -0.2679 r2: -3.7321 are real & different.

//P14) Grade , marks and average Caculator for given marks

```

import java.util.Scanner;

class GradeCalculator{

public static void main(String[] args){

int tel,eng,hin,maths,sc,soc,total_marks;

float avg;

Scanner s =new Scanner(System.in);

System.out.println("Enter telugu = ");

tel=s.nextInt();

System.out.println("Enter english = ");

eng=s.nextInt();

System.out.println("Enter hindi = ");

```

```
hin=s.nextInt();
System.out.println("Enter maths = ");
maths=s.nextInt();
System.out.println("Enter scince = ");
sc=s.nextInt();
System.out.println("Enter social = ");
soc=s.nextInt();
total_marks=(tel+eng+hin+maths+sc+soc);
avg=total_marks/6;
System.out.println("Total obtained marks : "+total_marks+"/600"+" \naverage = "+avg);
if(avg>=90 && avg<=100)
{
System.out.println("A+ Grade");
}
else if(avg>=80 && avg<90){
System.out.println("A Grade");
}
else if(avg>=70 && avg<80){
System.out.println("B Grade");
}
else if(avg>=60 && avg<70){
System.out.println("C Grade");
}
else if(avg>=44 && avg<60){
System.out.println("D Grade");
}
else if(avg>=0 && avg<44){
System.out.println("Fail");
}
else{
System.out.println("Invalid Marks");
}
}
```

```
}
```

output:

Enter telugu =

98

Enter english =

65

Enter hindi =

76

Enter maths =

86

Enter science =

45

Enter social =

88

Total obtained marks : 458/600

average = 76.0

B Grade

// P15) print tax, Gross salary and income from given salary from user

```
import java.util.Scanner;
```

```
public class SalaryTax{
```

```
public static void main(String[] args) {
```

```
double tax,salary,grossSal;
```

```
Scanner sc=new Scanner(System.in);
```

```
System.out.println("Enter Your Annual income: ");
```

```
salary=sc.nextDouble();
```

```
grossSal=salary;
```

```
if(salary>300000 && salary<=500000){
```

```
tax=salary*0.05; //5%
```

```
salary=salary-tax;
```

```
System.out.println("From your salary "+grossSal+"LPA 5% of tax "+tax+" is deducted.And you
```

```
received "+salary);
```

```
}  
else if(salary>500000 && salary<=700000){  
tax=salary*0.1; //10%  
salary=salary-tax;  
System.out.println("From your salary "+grossSal+"LPA 10% of tax "+tax+" is deducted.And you  
received "+salary);  
}  
else if(salary>700000 && salary<=1000000){  
tax=salary*0.15; //15%  
salary=salary-tax;  
System.out.println("From your salary "+grossSal+"LPA 15% of tax "+tax+" is deducted.And you  
received "+salary);  
}  
else if(salary>1000000 && salary<=1500000){  
tax=salary*0.2; //20%  
salary=salary-tax;  
System.out.println("From your salary "+grossSal+"LPA 20% of tax "+tax+" is deducted.And you  
received "+salary);  
}  
else if(salary>1500000 && salary<=2000000){  
tax=salary*0.25; //25%  
salary=salary-tax;  
System.out.println("From your salary "+grossSal+" LPA, 25% of tax "+tax+" is deducted.And you  
received "+salary);  
}  
else if(salary>2000000 && salary<=2500000){  
tax=salary*0.3; //30%  
salary=salary-tax;  
System.out.println("From your salary "+grossSal+" LPA, 30% of tax "+tax+" is deducted.And you  
received "+salary);  
}
```

```

else if(salary>2500000 && salary<=3000000){
tax=salary*0.35; //35%
salary=salary-tax;
System.out.println("From your salary "+grossSal+" LPA, 35% of tax "+tax+" is deducted.And you
received "+salary);
}
else if(salary>3000000){
tax=salary*0.40; //40%
salary=salary-tax;
System.out.println("From your salary "+grossSal+" LPA, 40% of tax "+tax+" is deducted.And you
received "+salary);
}
else {
System.out.println("No tax for less than 3LPA");
}
}
}
}

```

output1:

Enter Your Annual income:

1600000

From your salary 1600000.0 LPA, 25% of tax 400000.0 is deducted.And you received 1200000.0

output2:

Enter Your Annual income:

3200000

From your salary 3200000.0 LPA, 40% of tax 1280000.0 is deducted.And you received 1920000.0

//P16) Write a java program to print sum of digits in a given number

```

import java.util.Scanner;

class Main {

    public static void main(String[] args) {

        int num,sum=0;

```

```
Scanner s=new Scanner(System.in);
System.out.println("Enter a number: ");
num=s.nextInt();
while(num>0){
    int rem=num%10;
    sum+=rem;
    num=num/10;
}
System.out.println("Sum of digts for given number: "+sum);

}
}
```

output:

Enter a number:

4567

Sum of digts for given number: 22

//P17) print ascii value for given character

```
import java.util.Scanner;
class ChartoAscii{
    public static void main(String[] args){
        Scanner s=new Scanner(System.in);
        System.out.println("Enter a Character: ");
        char ch=s.next().charAt(0);
        int ascii_val=ch;
        System.out.println("Ascii value of "+ch+" is : "+ascii_val);
    }
}
```

output:

Enter a Character:

B

Ascii value of B is : 66

//P18)print the upper & lower case alphabets

```
class AtoZ
{
public static void main(String args[])
{
char ch ;
System.out.println("LowerCase Alphabets: ");
for(ch=97 ;ch<=122 ;ch++)
{
System.out.print(ch+" ");
}
System.out.println();
System.out.println("UpperCase Alphabets: ");
for(ch=65 ;ch<=90 ;ch++)
{
System.out.print(ch+" ");
}
}
}
```

output:

LowerCase Alphabets:

a b c d e f g h i j k l m n o p q r s t u v w x y z

UpperCase Alphabets:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

//P19) reverse the upper & lower case alphabets

```
class AtoZ
{
public static void main(String args[])
{
char ch ;
```



```

System.out.println("LowerCase Alphabets: ");
for(ch=122 ;ch>=97 ;ch--)
{
    System.out.print(ch+" ");
}

    System.out.println();
System.out.println("UpperCase Alphabets: ");
for(ch=90 ;ch>=65 ;ch--)
{
    System.out.print(ch+" ");
}
}
}

```

output:

LowerCase Alphabets:

z y x w v u t s r q p o n m l k j i h g f e d c b a

UpperCase Alphabets:

Z Y X W V U T S R Q P O N M L K J I H G F E D C B A

//P20) Write a java program to check the given character is vowel or consonant using switch case

```

import java.util.Scanner;

class VoweOrCon{

    public static void main(String[] args) {

        char ch;

        Scanner s=new Scanner(System.in);

        System.out.println("Enter a character: ");

        ch=s.next().charAt(0);

        if(ch<95){

            int c=ch+32;

            ch=(char)c;

        }

        switch(ch){

```

```
case 'a':  
    System.out.println("Vowel");  
    break;  
case 'e':  
    System.out.println("Vowel");  
    break;  
case 'i':  
    System.out.println("Vowel");  
    break;  
case 'o':  
    System.out.println("Vowel");  
    break;  
case 'u':  
    System.out.println("Vowel");  
    break;  
default:  
    System.out.println("Consonant");
```

```
}
```

```
}
```

```
}
```

output:

Enter a character:

p

Consonant

//21) Write a java program to check the given character is vowel or consonant using if-else statements

```
import java.util.Scanner;
```

```

class VoweOrCon{
    public static void main(String[] args) {
        char ch;
        Scanner s=new Scanner(System.in);
        System.out.println("Enter a character: ");
        ch=s.next().charAt(0);
        if(ch<95){
            int c=ch+32;
            ch=(char)c;
        }

        if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u'){
            System.out.println("Vowel");
        }
        else{
            System.out.println("Consonant");
        }

    }
}

```

OUTPUT:

Enter a character:

S

Consonant

//ATM

import java.util.*;

public class ATM2

{

```
public static void main(String args[] )
{

    int balance = 9000, withdraw, deposit;
    String pinSystem="0123",pinUser,temp;

    Scanner s = new Scanner(System.in);

    while(true)
    {
        System.out.println("<-- Automated Teller Machine -->\n");
        System.out.println("Choose 1 for Withdraw");
        System.out.println("Choose 2 for Deposit");
        System.out.println("Choose 3 for Check Balance");
        System.out.println("Choose 4 for Pin change");
        System.out.println("Choose 5 for EXIT");
        System.out.print("Choose the above options you want to perform: ");

        int choice = s.nextInt();

        switch(choice)
        {
            case 1:
                System.out.println("Enter Four digit Pin: ");
                pinUser=s.next();
                if(pinUser.length()==4 && pinSystem.equals(pinUser))
                {
                    System.out.print("Enter money to be withdrawn:");
                    withdraw = s.nextInt();
```

```

        if(withdraw%100==0 && withdraw<=10000){
            if(balance >= withdraw)
            {
                balance = balance - withdraw;
                System.out.println("Please collect your money....");
            }
            else
            {
                System.out.println("*** Insufficient Balance ***");
            }
        }
        else{
            System.out.println("Enter money in multiples of 100's or 200's or 500's Or enter valid
amount(>0 or <10000)\n");

        }

    }

}

else
{
    System.out.println("*** Invalid pin ***");
}

System.out.println("\n\n");

break;

case 2:

    System.out.println("Enter Four digit Pin: ");
    pinUser=s.next();
    if(pinUser.length()==4 && pinSystem.equals(pinUser))
    {
        System.out.print("Enter money to be deposited:");
    }

```

```
deposit = s.nextInt();
if(deposit%100==0){
    balance = balance + deposit;
    System.out.println("Your Money has been successfully deposite");
}
else{
    System.out.println("Your deposit "+deposit+" is unsuccessful..");
    System.out.println("Deposit money should be multiples of 100's or 500's");
}
}
else
{
    System.out.println("*** Invalid pin ***");
}
System.out.println("\n\n");

break;
```

case 3:

```
System.out.println("Enter Four digit Pin: ");
pinUser=s.next();
if(pinUser.length()==4 && pinSystem.equals(pinUser))
{
    System.out.println("Balance : "+balance);
}
else
{
    System.out.println("*** Invalid pin ***");
}
System.out.println("\n\n");

break;
```

case 4:

```
    System.out.println(" Reset Password ");
    System.out.print("Enter your old pin: ");
    temp=s.next();
    if(pinSystem.equals(temp)){
System.out.print("\nEnter new pin: ");
pinSystem=s.next();
        System.out.println("Pin Successfully changed...");
    }
    else{
        System.out.println("*** Invalid pin ***");
    }
    System.out.println("\n\n");
    break;
```

case 5:

```
    System.exit(0);
}
}
}
}
```

//Basic calculator operations using switch case

```
import java.util.*;
class Calculator
{
    public static void main(String args[])
    {
        System.out.println("Ente two number = ");
        Scanner sc = new Scanner(System.in);
```

```
double first = sc.nextDouble();
double second = sc.nextDouble();
double result ;
System.out.println("Select any operator (+ , - , * , / , %)");
char operator = sc.next().charAt(0);

switch(operator)
{
case '+':
result = first + second ;
System.out.println(first+" "+operator+" "+second+"="+" "+result);
break;
case '-':
result = first - second ;
System.out.println(first+" "+operator+" "+second+"="+" "+result);
break;
case '*':
result = first * second ;
System.out.println(first+" "+operator+" "+second+"="+" "+result);
break;
case '/':
result = first / second ;
System.out.println(first+" "+operator+" "+second+"="+" "+result);
break;
case '%':
result = first % second ;
System.out.println(first+" "+operator+" "+second+"="+" "+result);
break;

default:
System.out.println("Error");
```



```
}
```

```
}
```

```
}
```

//Prime numbers in the first 20 Fibonacci series

```
class PrimeFromFibonacci{  
    public static void main(String[] args) {  
        int a=0,b=1,temp,count;  
        for(int i=1;i<=20;i++){  
            count=0;  
            for(int j=1;j<=a;j++){  
                if(a%j==0){  
                    count++;  
                }  
            }  
            if(count==2){  
                System.out.println(a);  
            }  
            temp=a+b;  
            a=b;  
            b=temp;  
        }  
    }  
}
```

//number to word in between 10 & 99

```
import java.util.Scanner;  
  
class NumToWord{  
    public static void main(String[] args){  
        int rev=0;  
        Scanner s=new Scanner(System.in);
```

```
System.out.println("Enter number between 10 & 99: ");
int num=s.nextInt();
while(num>0){
    int rem=num%10;
    rev=rev*10+rem;
    num=num/10;
}

for(int i=1;i<=2;i++){
    int num2=rev%10;
    rev=rev/10;
    switch(num2){
        case 1:
            System.out.print("one ");
            break;
        case 2:
            System.out.print("Two ");
            break;
        case 3:
            System.out.print("Three ");
            break;
        case 4:
            System.out.print("Four ");
            break;
        case 5:
            System.out.print("Five ");
            break;
        case 6:
            System.out.print("Six ");
            break;
        case 7:
```

```
        System.out.print("Seven ");
        break;
    case 8:
        System.out.print("Eight ");
        break;
    case 9:
        System.out.print("Nine ");
        break;
    case 0:
        System.out.print("Zero ");
        break;
    }
}
}
```